



MAJI HYDROPOWER PROJECT

Cost-Benefit Analysis Summary Report

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1. Introduction and Objective

The Maji Hydro Project is a proposed grid-connected hydropower project intended to supply reliable electricity to both public and private users. Developed under a Public-Private Partnership (PPP) structure, the project is designed to displace high-cost diesel generation, enhance rural electrification, and contribute to broader environmental and developmental objectives.

This report provides an integrated assessment of the project's financial feasibility, economic viability, and distributional implications. It employs a cost-benefit analysis framework grounded in economic pricing principles – drawing on shadow pricing, conversion factors, and opportunity cost measures. The evaluation further examines how benefits and costs are distributed among key stakeholders, including government, labour, utilities, and local communities. The report concludes by identifying the critical risk parameters affecting project outcomes and outlining the conditions under which the project remains justified from both financial and economic perspectives.

2. Financial Appraisal

The financial performance of Maji Hydro Project was assessed across four policy scenarios, varying in tariff indexation and the duration of the corporate income tax holiday. The analysis focuses on the project's return to equity holders, the bankability of the financing structure, and the robustness of debt service capacity. Under the base case scenario (non-indexed tariffs and a 10-year tax holiday), the project yields an Equity IRR of 15.4% and an NPV of USD 0.82 million. The minimum Annual Debt Service Coverage Ratio (ADSCR) is 1.54, with an average of 2.08, indicating a modest but acceptable level of cash flow coverage for lenders. Under current assumptions, the project is likely to meet its debt obligations on an annual basis, though the financing structure would benefit from further stress testing under adverse conditions. However, at the project level, the pre-financing NPV is USD -4.97 million, suggesting the project is not financially sustainable on purely commercial terms without fiscal incentives or concessional structuring.

Financial viability improves considerably when tariffs are indexed to inflation. In the scenario with tariff adjustment and a 10-year tax holiday, the Equity IRR increases to 20.35% and the Equity NPV rises to USD 4.24 million, with coverage ratios strengthening accordingly. Conversely, reducing the tax holiday from 10 to 5 years materially weakens project performance. Without tariff indexation, a 5-year holiday results in a negative Equity NPV of USD -0.47 million and an Equity IRR of 13.14%, with the ADSCR declining to 1.36. The scenario combining tariff adjustment with a shorter 5-year tax holiday represents an

intermediate scenario, with an Equity IRR of 18.02% and an NPV of USD 2.55 million. While still financially viable, this case reflects the importance of balancing tariff structures with fiscal incentives to ensure both financial viability and sustained investor confidence.

From a financing perspective, while the project's debt structures – comprising multilateral debt, commercial borrowing, and 40 percent equity – are serviceable under most policy combinations, the broader results underscore the sensitivity of investor returns to tariff escalation clauses and tax policy commitments, both of which play a pivotal role in ensuring the bankability of the project. These financial findings also serve as a foundation for understanding the distributional impact on equity holders and fiscal authorities, as elaborated in the following sections.

3. Economic Appraisal

The project's economic viability was assessed using shadow pricing and conversion factors to account for market distortions in taxes, subsidies, and trade protection. The analysis evaluates whether the project enhances national welfare by comparing its resource cost savings and positive externalities against the economic opportunity cost of capital (EOCK).

Under the base case scenario, the project yields an Economic IRR of 25.3% and Economic NPV of USD 17.66 million, discounted at the EOCK of 11.5%. This indicates that the project is economically viable and represents efficient use of national resources, particularly through the substitution of high-cost thermal generation and delivery of rural electricity access. Across the alternative policy scenarios, the project maintains robust economic returns. When tariffs are indexed and the 10-year holiday is retained (Case 2), the Economic IRR falls to 22% and the Economic NPV declines to USD 12.99 million. The slight reduction reflects the increased cost to the economy from higher real payments under the inflation-adjusted tariff, which shifts economic surplus from consumers to investors through higher real payments.

Conversely, removing the tax holiday while keeping tariffs non-indexed (Case 3) results in a higher Economic IRR at 26% and an Economic NPV of USD 19.36 million. This increase is driven by earlier tax inflows to the government, which enhance overall economic returns despite a decline in financial attractiveness. In the combined case of tariff indexation and a shortened 5-year holiday (Case 4), the project yields an Economic IRR of 24% and an Economic NPV of USD 15.23 million. While slightly below the base case, the outcome remains well above the EOCK threshold, reaffirming the project's economic merit.

Overall, the economic results demonstrate that the Maji Hydro Project delivers substantial welfare gains under all policy configurations. The strong positive NPVs and IRRs

above the opportunity cost of capital validate the project's contributions to efficient resource allocation, even in scenarios where financial returns are constrained. This underscores the justification for public support in improving private viability where warranted. This reinforces the case of prioritizing concessional finance and regulatory alignment as tools to translate economic value into financially bankable outcomes.

4. Stakeholder and Distributional Impact

The distributive analysis yields a NPV Externality of USD 2.94 million in the base case, indicating that the project generates a net economic surplus for domestic stakeholders. This positive outcome is largely driven by substantial gains to the Utility (USD 9.01 million) and Selected Communities (USD 0.59 million), which outweigh the losses incurred by Labour (USD -2.31 million) and the Government (USD -4.35 million). The NPV Stakeholder of USD 21.61 million further reflects welfare gains due to corrections for market distortions.

Across alternative policy configurations, this favourable aggregate trend persists. In Case 2, the Stakeholder NPV remains positive at USD 16.81 million, though Government losses deepen to USD -5.88 million and Utility gains decline to USD 2.95 million due to inflation-indexed payments. In Case 3, the Stakeholder NPV peaks at USD 23.31 million, benefiting from both early tax inflows and strong returns to the utility (USD 9.12 million). In Case 4, the NPV remains robust at USD 19.05 million, despite the Government facing its highest loss (USD -7.99 million).

The consistent finding is that while the aggregate Stakeholder NPVs are all positive, distributional asymmetries persist. Government and labour consistently bear economic costs, even under economically viable configurations. This underlines the importance of fiscal redistribution, tariff structure calibration, and targeted compensatory mechanisms to ensure political acceptability and equitable benefit sharing. Without such balancing instruments, there is a risk that long-term domestic support may weaken, even for projects that are otherwise socially desirable.

5. Sensitivity and Risk Analysis

The project's outcomes are highly sensitive to changes in key assumptions. For instance, a 15% drop in PPA tariffs would reduce equity NPV from USD 0.82 million to USD -3.26 million, pushing the project close to financial non-viability. Conversely, a 15% tariff increase raises equity NPV to USD 4.9 million, highlighting how even modest tariff adjustment lead to a large swings in equity NPV, underscoring high price elasticity of inventor returns. While IPPs request corporate tax holidays, the 5-year tax holiday lowers government revenues and reduces equity NPV to USD -3.4 million, without offering broader

resilience. Capital cost overruns of +20% drop NPV to USD 0.04 million, while 20% O&M cost hike reduces it to USD -1.7 million. On the economic side, higher fuel prices (+10%) improves economic NPV to USD 22.18 million, while a reduction in capacity utilization to 65% lowers it to USD 15.33 million.

These findings confirms the project is not only viable but also financially resilient under base conditions, while still vulnerable to adverse shifts in key assumptions. Risk-sharing must be carefully designed: protecting private returns without compromising fiscal sustainability or the equitable distribution of national benefits. These stress-test results reveal that while the project is fundamentally strong, its long-term sustainability hinges on realistic assumptions regarding macroeconomic risks, cost structures, and tariff mechanisms – factors essential for robust financial structuring and fair stakeholder outcomes.

6. Conclusion and Recommendations

The Maji Hydropower Project demonstrates robust economic viability across all policy scenarios, with Economic IRRs consistently exceeding the EOCK. However, its financial sustainability is contingent upon the inclusion of tariff indexation mechanisms and continued fiscal concessions such as tax holidays. The distributive analysis highlights that while domestic stakeholders collectively benefit in net terms (USD 21.61 million), this surplus is unevenly distributed, with Government and Labour still incurring net losses.. It is recommended that targeted compensatory mechanisms, such as performance-based subsidies or strategic reallocation of fiscal revenues, be explored to enhance distributive equity. Sensitivity results further emphasize tariff levels and cost control as critical risk parameters. To secure both financial bankability and long-term social acceptability, regulatory clarity and equitable risk-sharing arrangements must be strengthened.